

Capstone Project - 01

Rule Engine Implementation

Acceptance Criteria: Rules shall vary from client to client hence the Rules have to be made configurable. The Rule Engine should be able to pick up these rules, execute them and assign an end state for the records.

Rule Execution End States

Sequence	Description	Status
1	Bank Transaction uploaded and not matched with Open AR/Remittance	Open
2	Bank Statement & Open AR/Remittance Value matched with variance	Partial Match
3	Bank Statement & Open AR/Remittance Value fully matched	Closed
4	No Invoice details in Bank Statement & No Remittance Advice	UAC (Un-applied Cash)
5	No information in Bank Statement	UIC (Un-Identified Cash)
6	User queried for more information	Query

Rules

The Rules are primarily defined on the columns from Bank Statement, Open AR and Remittance Advices configured for each deal.

Below are few Sample rules:

Priority	Bank Statement	ERP AR Open	Remittance Advice	Match	Partial Match
1	Customer Name+PO number+Amount	Customer Name+PO number (get Invoice number from it) + Amount		2 Way	Unmatched value after deduction should be posted into UAC
2	Customer Name+Delivery number+Amount	Customer Name+Delivery number (get Invoice number from it)+Amount		2 Way	Unmatched value after deduction should be posted into UAC
3	Customer Name+Invoice number+Invoice Date+Amount	Customer Name+Invoice number+Invoice Date+Amount		2 Way	Unmatched value after deduction should be posted into UAC
4	Customer Name+Invoice number+Amount	Customer Name+Invoice number+Amount		2 Way	Unmatched value after deduction should be posted into UAC
5	Customer Bank Payment	Customer Name+PO number+Invoice number+Invoice Date+Amount	Customer Name+PO number+Invoice number+Invoice Date+Amount	3Way	Unmatched value after deduction should be posted into UAC
6	Customer Bank Payment	Customer Name+PO number+Invoice number+Amount	Customer Name+PO number+Invoice number+Amount	3Way	Unmatched value after deduction should be posted into UAC

- **Priority 1 Rule:** The Customer Name, PO Number is available, get invoice number mapped to that PO number and the Invoice Amount using the **Bank statement** where it mentioned has to be compared with **Open AR** Customer Name, Invoice Number (mapped to that PO Number) and the Invoice Amount.
- **Priority 2 Rule:** The Customer Name, Delivery Number, Invoice Number and the Invoice Amount from the **Bank statement** has to be compared with **Open AR** Customer Name, Invoice Number (mapped to delivery number) and the Invoice Amount.
- **Priority 3 Rule:** The Customer Name, Invoice Number, Invoice Date and the Invoice Amount from the **Bank statement** has to be compared with **Open AR** Customer Name, Invoice Number, Invoice Date and the Invoice Amount.
- **Priority 4 Rule:** The Customer Name, Invoice Number and the Invoice Amount from the **Bank statement** has to be compared with **Open AR** Customer Name, Invoice Number and the Invoice Amount.
- **Priority 5 Rule:** The Customer Name, PO Number, Invoice Number, Invoice Date and the Invoice Amount from the **Remittance Advice** has to be compared with **Open AR** Customer Name, PO Number, Invoice Number, Invoice Date and the Invoice Amount.

- **Priority 6 Rule:** The Customer Name, PO Number, Invoice Number and the Invoice Amount from the **Remittance Advice** has to be compared with **Open AR** Customer Name, PO Number, Invoice Number and the Invoice Amount.